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Multibit, Lead-Free Cs_2SnI_6 Resistive Random Access Memory with Self-Compliance for Improved Accuracy in Binary Neural Network Application

Ajit Kumar, Mokurala Krishnaiah, Jinwoo Park, Dhananjay Mishra, Bidyashakti Dash, Hyeon-Bin Jo, Geun Lee, Sangwook Youn, Hyungjin Kim, and Sung Hun Jin**

In the realm of neuromorphic computing, integrating Binary Neural Networks (BNN) with non-volatile memory based on emerging materials can be a promising avenue for introducing novel functionalities. This study underscores the viability of lead-free, air-stable Cs_2SnI_6 (CSI) based resistive random access memory (RRAM) devices as synaptic weights in neuromorphic architectures, specifically for BNNs applications. Herein, hydrothermally synthesized CSI perovskites are explored as a resistive layer in RRAM devices either on the rigid or flexible substrate, highlighting reproducible multibit switching with self-compliance, low- resistance-state (LRS) variations, a decent On/Off ratio(or retention) of ≈10^3 (or 10^4 s), and endurance exceeding 300 cycles. Moreover, a comprehensive evaluation with the 32 × 32 × 3 RGB CIFAR-10 dataset reveals that binary convolutional neural networks (BCNN) trained solely on binary weight values can achieve competitive rates of accuracy comparable to those of their analog weight counterparts. These findings highlight the dominance of the LRS for CSI RRAM with self-compliance in a weighted configuration and minimal influence of the high resistance state despite substantial fluctuations for flexible CSI RRAM under varying bending radii. With its unique electrical switching capabilities, the CSI RRAM is highly anticipated to emerge as a promising candidate for embedded AI systems, especially in IoT devices and wearables.

perovskite(ABX3) pb-based
-> LFDP(lead-free double inorganic perovskites(A2BX6) :CS2SnI6(CSI)
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1. Introduction

In the rapidly evolving landscape of information technology, the quest for memory devices that combine high performance, energy efficiency, and novel functionalities has intensified. Among the diverse range of non-volatile memory types, resistive random-access memory (RRAM) has garnered significant attention as it offers important advantages of scalability (≈$4F^2$), speed (≈100 ps),[1] power consumption (≈0.06/0.9 pJ bit^{-1}),[2] and endurance (≈10^{12}),[3] among others.[4] However, these devices also face challenges related to variability, data retention, write disturbances, and multilevel cell stability, to list a few.[5] In this sense, the pursuit of RRAM materials that promise higher performance, better energy efficiency, and novel functionalities remains incessant.[6] Accordingly, an emerging class of materials such as perovskites has attracted intense interest given their potential, such as higher On/Off ratios, reduced switching voltages, unique optical multiple functionalities, and compatibility with novel flexible form factors, among others.[7,8] However, Pb-based materials, which are hazardous and not highly stable in the air, represent the major limitations for most emerging perovskites with regard to adopting RRAM materials in practical application.[9] As an alternative to Pb-based perovskites, lead-free double inorganic perovskites (LFDP) exhibit structural stability under various ambient conditions, including environmental (moisture), thermal, and light exposure conditions. Among the diverse range of LFDPs currently available, Cs_2SnI_6 (CSI) has been reported as a promising LFDP candidate for photodetectors (PDs),[9] solar cells,[10] water-splitting applications[11] and thermistors[12] owing to its structural stability in air, thermal, and light conditions.[13] Interestingly, the resistive switching (RS) characteristics of the lead-free CSI perovskite layer and corresponding memory applications are rarely explored in the literature despite the significant capability of this material to provide a pivotal advancement in RRAM technology.

In this study, we investigated the possible use of CSI RRAM as an inorganic lead-free perovskite given the following

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performance metrics: i) self-compliance properties, eliminating the need for an external compliance current, facilitating controlled switching, and reducing the risk of catastrophic failures, ii) multibit storage capabilities, enabling significant enhancement of memory density, iii) lower switching voltages, leading to reduced power consumption, and iv) flexible form factors, among other merits. Given these traits, the hydrothermally synthesized CSI particles could be converted to a polycrystalline thin film via a drop-casting technique. The fabricated RRAM devices exhibited reproducible bipolar RS with good self-compliance, an On/Off ratio of $\approx 10^3$, excellent retention of 10^4 s, endurance exceeding 300 cycles, multi-bit (6-level, reset-based) operation, low LRS variations, and decent device uniformity. However, the origin of self-compliance behaviors has rarely been addressed for the material pool of perovskites, despite the fact that it may be a core issue for high-density and low-cost cross-bar RRAM applications.

Herein, the origin of self-compliance in CSI RRAM materials is comprehensively investigated and compared with several reported conventional materials,[14,15] including metal oxides[16] and others.[14,15,17] In addition, the low variation of the low resistance state (LRS) for CSI RRAM, stemming from the self-compliance effects, is experimentally highlighted. Further, this property is validated in practice to contribute to advancing binary neural networks (BNNs), a subset of neural networks that utilize binary values for computations, significantly reducing energy consumption levels. The integration of perovskite-based resistive multi-bit memory with self-compliance and a BNN is expected to represent a significant advancement in-memory technologies and artificial intelligence.

2. Results and Discussion

2.1. Physical Properties of CSI Particles and Films

Figure 1 presents the background of this work, highlighting the motivation, key requirements for BNN applications, and CSI synthesis protocols. Research on accelerating deep neural network (DNN) computation is receiving increased attention, especially as DNNs exhibit exceptional performance across various artificial intelligence disciplines.[18] A primary challenge in contemporary DNN acceleration lies in the extensive data movement between on-chip processors and off-chip memory.[19] In contrast, the binary neural network (BNN) is garnering interest due to its significantly reduced memory requirements (Figure 1a-i). For a highly integrated BNN hardware system, an RRAM cross-bar array (RCA) is particularly suitable, boosting the potential to achieve the most compact design with an area as small as $4F^2$.[20] However, as shown in the diagram (Figure 1a-ii), the accuracy of BNNs can be significantly compromised by variations in the RRAM devices, be it device-to-device or cycle-to-cycle. Lead-free double inorganic perovskite particles, specifically CSI, have been synthesized in the hydrothermal process, as depicted in Figure 1a-iii. These CSI devices, characterized by their trap control and non-stoichiometric behavior, have demonstrated the potential to enhance BNN accuracy due to their low-LRS variation and high On/Off ratio.

The hydrothermal synthesis details of the CSI particles are elaborated upon in the Experimental Section. X-ray powder diffraction (XRD) confirmed the cubic crystal phase of the CSI particles[9] (Figure S1, Supporting Information). The crystal phase, relevant to structural stability, for the hydrothermally synthesized CSI particles was stable over three months in the air (Figure S1a, Supporting Information) and one year in the glove box (Figure S1b, Supporting Information). The structural stability of CSI particles depends on processing techniques and their parameters (for example, solvents and initial precursors, Table S1, Supporting Information).[9] The structural stability of CSI particles synthesized via different process techniques is summarized in Table S1 (Supporting Information). The hydrothermally synthesized CSI particles with diluted hydriodic acid (HI) solvents exhibit enhanced structural stability in air and glove box compared to other synthesis processes (Table S1, Supporting Information). According to density functional/hybrid density functional theory (DFT/HDFT), [21] the structural stability of lead-free CSI particles is ascribed to the strong covalency of the Sn–I bond in the isolated $[SnI_6]^{2-}$ cluster and the presence of the higher oxidation state (+4) of Sn in CSI compound.[21] The field emission scanning electron microscopy (FE-SEM) revealed the octahedral morphology of the particles, consistent with previously reported literature[11] (Figure S2, Supporting Information). The energy-dispersive X-ray (EDS) spectrum estimated the average elemental composition (atomic (%)) of the CSI particles as Cs at (20.60 ± 0.34)%, Sn at (11.80 ± 0.46)%, and I at (67.60 ± 0.56)% (Figure S3 and Table S2, Supporting Information). Bright-field emission transmission electron microscopy (FE-TEM) showed that the CSI nanoparticles possess a cubical-octahedral shape, ranging from 400 to 500 nm in size (Figure S4, Supporting Information). A selected area diffraction of a single particle displayed dot patterns, suggesting that the CSI particle has a single-crystalline structure (Figure S5, Supporting Information). High-resolution TEM (HR-TEM) images showed distinct lattice fringes, indicating the crystallinity of the particles (Figure S5, Supporting Information). HR-TEM and FFT analyses identified interplanar spacings of d = 0.33 and 0.21 nm, corresponding to lattice planes (222) and (022), respectively (Figure S5, Supporting Information). These planes align with the cubic phase of the CSI particles. HR-TEM and XRD analyses support the high crystallinity of the synthesized CSI particles. Lastly, high-angle annular dark-field (HAADF)-EDS elemental mapping affirmed the homogeneous distribution of Cs, Sn, and I within individual CSI particles (Figure S6, Supporting Information).

Figure 1b displays the schematic diagram of the RRAM structure (ITO/CSI/Ag), while Figure 1c showcases the corresponding cross sectional FE-SEM images of the devices. A comprehensive account of the CSI-based RRAM device fabrication process can be found in the Experimental Section (Sections 4.2 and 4.3). The RRAM device is structured with ITO as the bottom electrode (BE), CSI films (formed by drop casting) as the RS layer, and Ag serving as the top electrode (TE). The cross sectional FE-SEM images highlight a dense and compact adherence between the CSI RS layer and the ITO (BE) interface (Figure 1c). The thickness of the CSI RS layer is ≈700 nm, as determined from the microscopy (Figure 1c). Interestingly, the surface of the drop-casted CSI RS layer (Figure 1d), which is dense and compact, presents a stark contrast to hydrothermally synthesized particles (Figure S2, Supporting Information). Elemental mapping emphasizes the consistent distribution of the elements Cs, Sn, and I over a chosen region of the drop-casted CSI RS layer

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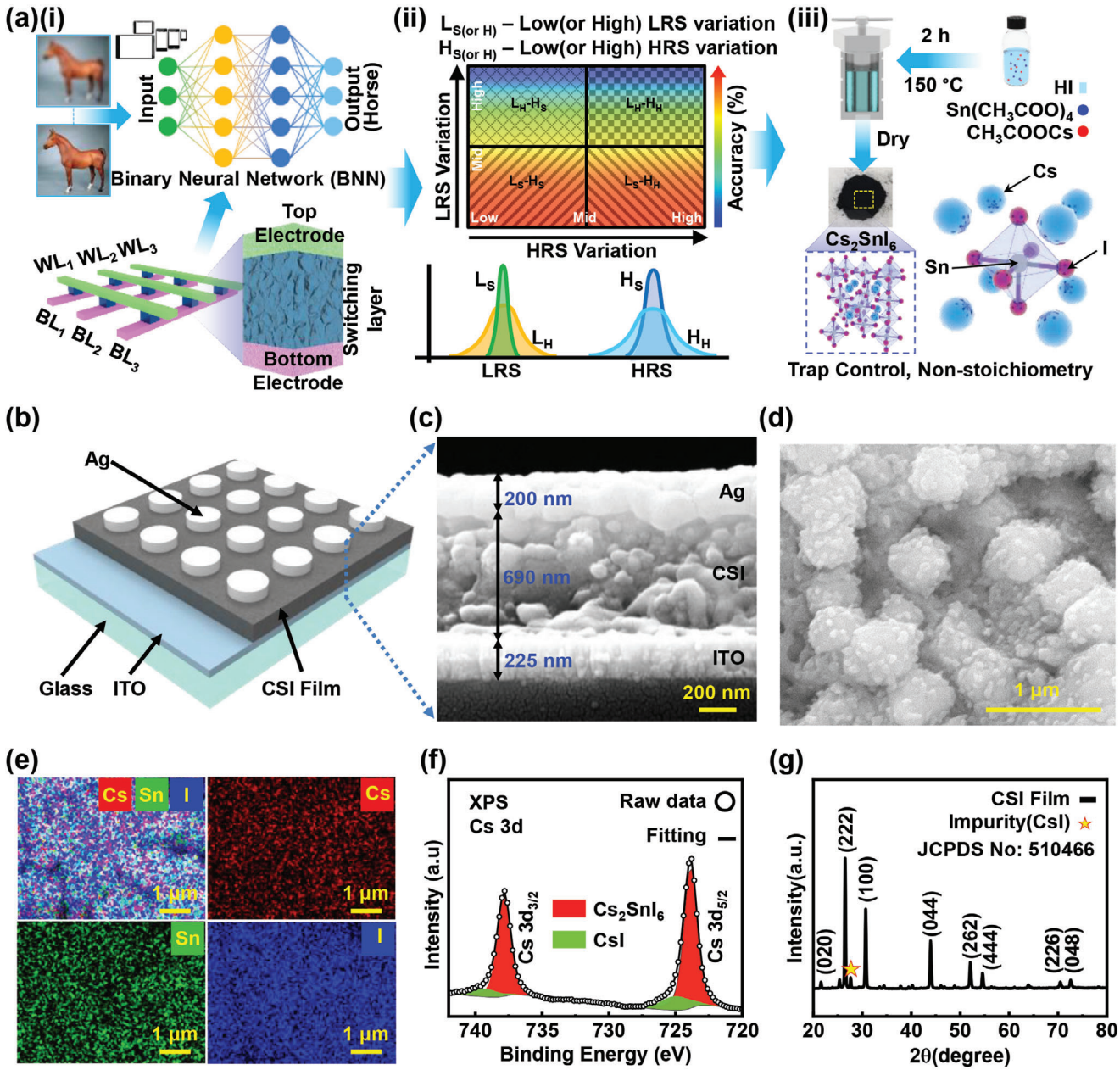


Figure 1. CSI RRAM: BCNN Conceptualization with CSI RRAM and its physical characterization a-i) Depiction of cross-bar RRAM devices tailored for BCNN applications, ii) Analysis of the impact of LRS-HRS fluctuations on the accuracy. iii) Sequential process flow of the lead-free inorganic CSI perovskite particles via hydrothermal process. b) Schematic diagram of the CSI RRAM device. c) Cross Sectional FE-SEM image of the crafted CSI device. d) Surface SEM imaging of the drop casted CSI film and e) corresponding EDS mapping of constituent elements. f) Core level XPS spectra of Cs3d and (g) X-ray diffraction pattern of drop casted CSI films.

(Figure 1e; Figure S7, Supporting Information). The EDS spectrum indicates that the average elemental composition (atomic (%)) of the produced films (Table S3, Supporting Information) is skewed compared to the synthesized particles (Table S2, Supporting Information). The film has Cs-rich (24.15 ± 0.54)%, Sn-rich (13.18 ± 0.32)%, and I-deficient (62.67 ± 0.87)%, whereas the compositions of the synthesized particles contrast notably with increased Cs(+3.55%), Sn (+1.38%) (or reduced, I (−4.93%)) atomic % variation. The previous research suggests that process parameters like solvent choice,[9,19,22] the initial concentration of CSI particles in the solvent, and CSI paste-making procedures, including ultrasonication, magnetic stirring, and film-processing techniques,[22] impact the morphology of film and its elemental composition.[9,19,22–24] In this study, the solvent and procedures involved in CSI paste-making and annealing treatments could be responsible for the observed alterations in morphology and chemical composition. Broad scanning survey XPS spectra of the CSI RS layer confirm the presence of the constituents Cs, Sn,

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I, and a reference carbon element (Figure S8a,b, Supporting Information). High-resolution core-level spectra of Cs 3d, Sn 3d, and I 3d have been split into two peaks each (Figure 1f; Figure S8c,d, Supporting Information), providing insight into their respective energy values and confirming the presence of certain valence states in the CSI RS layer.[9] A supplementary peak in the Cs 3d spectra suggests the presence of CsI (Figure 1f).[13] In the XRD pattern of the drop-casted CSI RS layer, all diffraction peaks align well with the cubic crystal structure of CSI (JCPDS No: 510 466), but there is a minor discrepancy ≈25.6° (Figure 1g). The stray peak ≈26.5° is attributed to CsI (JCPDS No: 06–0311), a similar impurity previously observed during the conversion of particles to CSI films.[22]

2.2. RRAM Application

The fabricated CSI RRAM device exhibits a high resistance, exceeding 1 M Ω (Figure S9, Supporting Information). This necessitates an electroforming process to initiate the RS property of the device.[25–27] Electroforming was achieved by applying a negative voltage sweep (0 V → -5 V → 0 V) to the top Ag electrode, maintaining a current compliance (CC) of 1 mA (Figure S10, Supporting Information). A noticeable surge in current occurred after −3.8 V, transitioning the device to an LRS. This threshold voltage is called the forming voltage (see Figure S10, Supporting Information). The RS behavior of the CSI device was assessed by applying a voltage sweep of 0 → −1.5 → 0 → 3 → 0 V to the top Ag electrode. The recorded *I–V* curves demonstrate the bipolar RS nature of the crafted CSI-based devices (**Figure 2a**). When the device underwent a negative voltage sweep (0 V → −1.5 V → 0 V) with the ITO (BE) electrode grounded, a pronounced increase in current was detected ≈−0.82 V. This indicates the RRAM device's transition from the high resistive state (HRS) to LRS, a transition termed the set process. The corresponding voltage for this transition is the set voltage, established at −0.82 V. The electric field generated during the set process was $\approx 1.2 \times 10^4$ Vcm^{-1} for the CSI film, with a thickness of 690 nm. Switching the device back to HRS occurs during a positive voltage sweep (0 V → 3.0 V → 0 V), a process termed "reset," with the reset voltage (V_{reset}) being 1.94 V. The gradual current reduction during this reset indicates potential multilevel RS properties in CSI devices. Moreover, the observed non-linear I-V curves highlight the self-compliance characteristics (SCCs) of the CSI RRAM (Figure 2a). This property can aid in crafting a reliable RRAM circuit design, eliminating the need for CC during RS operations. In previous studies, SCCs in RRAM devices with lead-based metal halide perovskite RS layers were attributed to the inclusion of interfacial layers[28] and the existence of defects such as iodine vacancy (V_I), iodine interstitials (I_i) and others.[29–32] Nonetheless, a comprehensive understanding of SCCs in metal halide perovskite-based RRAM remains relatively unexplored or scarcely documented in the literature.

In oxide-based RRAM, various factors influence SCCs. These include the thickness of the material,[15] non-stoichiometric chemical composition, notably oxygen concentration,[16] defect concentration (like oxygen vacancies),[14,15] intrinsic defects, the strategic control over 1D or spatially restricted conducting filament growth seen in materials like CuO and ZnO nanowires,[14,17] physical parameters such as the dielectric constant[15] and thermal conductivity,[33] the presence of a metallic phase or the doping with metal ions,[17] and various process parameters.[14] Such process parameters include annealing temperature and atmosphere,[16] the material and thickness of the top electrode,[34,35] the area of the top electrode pad,[36] bilayer device configuration, and controlled joule heating. Additionally, the formation of an interfacial layer at the interface of the top electrode and RS layer has also been understood to be a significant factor. In this study, the observed SCCs in the CSI device could be linked to the non-stoichiometric chemical composition of the CSI RS layer, particularly the presence of iodine vacancy (V_I) or potentially due to a minor impurity phase, such as CsI. To better understand its contribution to SCCs, we probed the initial resistance and RS properties of the impurity phase in the ITO/CsI/Ag configuration. The acquired *I–V* characteristics reveal a notably high initial resistance and SCCs of the ITO/CsI/Ag device, as depicted in (Figure S11, Supporting Information). This impurity phase (CsI) could potentially serve as a series resistor in the device configuration, thus leading to the observed SCCs in CSI RRAMs. Moreover, SCCs were identified in CSI RRAMs made with both active top electrodes (like Ag, Ti, and Cr) and inert ones (such as Au) (Figure S12, Supporting Information), effectively ruling out the role of interfacial defects between the RS layer and the top electrode in contributing to SCCs. To verify the repeatability of the RS traits of the CSI RRAM devices, endurance and retention tests were conducted. The results, displaying over 300 *I–V* switching cycles (from the endurance test), are presented in Figure 2b. The derived HRS and LRS states from this endurance test, at a read voltage (V_{read}) of ±0.3 V, can be observed in Figure 2c. The displayed endurance traits indicate a consistent switch between HRS and LRS across more than 300 cycles, boasting an On/Off ratio of $(1.55 \pm 1.05) \times 10^3$.

Figure 2d plots the statistical distribution of the extracted SET/RESET voltages across these switching cycles. The extracted SET and RESET voltage statistics are – (0.82 ± 0.16) V and (1.94 ± 0.54) V, respectively, with coefficients of variation (CV) of 20% and 28%. This indicates a well-defined switching voltage for both SET and RESET processes. The *I–V* characteristics of 100 devices were analyzed to evaluate variation across devices in RS properties (Figure S13, Supporting Information). The calculated average values for LRS and HRS are (37 ± 3.6) Ω and $(3.6 \times 10^4 \pm 5.4 \times 10^4)$ Ω with CVs of 7.76% and 148.04%, respectively. Notably, the CSI-based RRAM device showcases a reduced CV percentage for the LRS, possibly due to the SCCs of the device. The diminished CV in LRS is consistent with findings from most oxide and halide perovskite-based RRAMs with SCCs (Table S4, SI). The previous studies indicate that variability in LRS or HRS can impact the BNN accuracy.[37,38] In this study, the minimal LRS variation (CV ≈7.76%) suggests that CSI-based RRAM devices could be highly beneficial for BNN applications.[38] The average On/Off ratio across the 100 devices is $\approx 10^3$ (Figure 2e), with corresponding set/reset voltages illustrated in Figure 2f. Significant deviations in the On/Off ratio and SET/RESET voltages were observed across the 100 devices, likely due to inconsistencies in the thickness of the drop-coated CSI films. Exploration of the gradual reset process during switching cycles highlights the potential for multilevel resistive states in CSI RRAM. Literature indicates that multiple resistive states in RRAM devices

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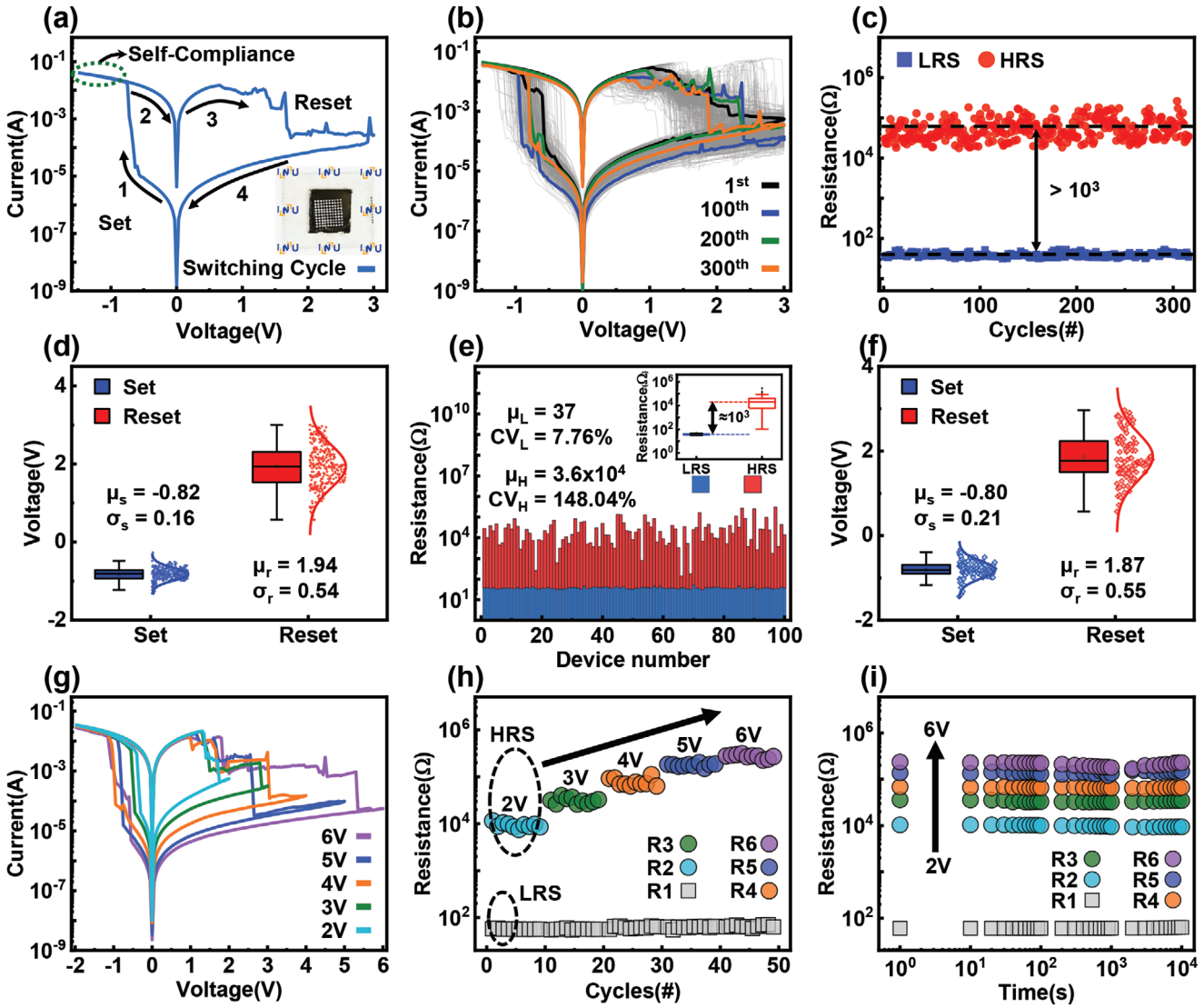


Figure 2. Resistive switching characteristics of CSI RRAM device, device uniformity, and multi-bit memory a) self-compliance, bipolar I-V RS characteristic of CSI device. b) Endurance of the device over 300 switching cycles. c) HRS and LRS derived at $V_{read} = \pm 0.3$ V and d) statistical distribution of set/reset voltages determined from endurance analysis. Device uniformity is confirmed by measuring the RS characteristics of 100 CSI device and their corresponding e) HRS/LRS states at $V_{read} = \pm 0.3$ V and f) statistical distribution of set/reset voltage devices, respectively. g) Reset voltage-dependent multi–level *I–V* RS characteristics CSI device and their corresponding h) endurance and i) 10^4 s retention analysis.

can be realized through methods like modulating current compliance during the set process,[28,39,40] adjusting the reset stop voltage in the reset process,[28,32,41] using light illumination,[39,42] and tweaking pulse width.[5,27] In many studies, the multilevel resistance states in lead-free perovskite memory devices have been manipulated by adjusting the current compliance during the set process.[31,43–48] In our study, multiple HRS states (R2-R6) were achieved by incrementing the reset stop voltage from 2 to 6 V at intervals of 1 V (Figure 2g). A 10-cycle endurance test showcases consistent switching and clear distinctions between the multiple HRS states (Figure 2h; Figure S14, Supporting Information). Retention tests of the six resistance states (R1-R6) over 10^4 s validate the stability of each state (Figure 2i; Figure S15, Supporting Information). This research underscores the ability for multilevel (i.e., six states) operation in the current self-compliant CSI RRAM device. Possible causes for the observed reset-based multiple HRS states might include defect migration and recombination within the RS layer,[32,41,49] defects at the RS layer/electrode interface,[41] differences in filament rupture,[50] and filament dimensions.[29] A comprehensive discussion is provided in the charge transport and conduction mechanism section. The multilevel RS observed in the ITO/CSI/Ag memory device suggests potential utility in multi-bit data storage for emerging computing systems.[51] Moreover, this multilevel storage capability, combined with self-compliance, accentuates a crucial memory attribute of RRAM, simplifying circuit complexity and memory scheme. To investigate the carrier transport and elucidate the conduction mechanism of ITO/CSI/Ag-based RRAM devices, we plotted the linear fitting of the log(I)-log(V) curves for both the SET and reset processes in **Figure** 3a,b. When the data is fitted with a slope of

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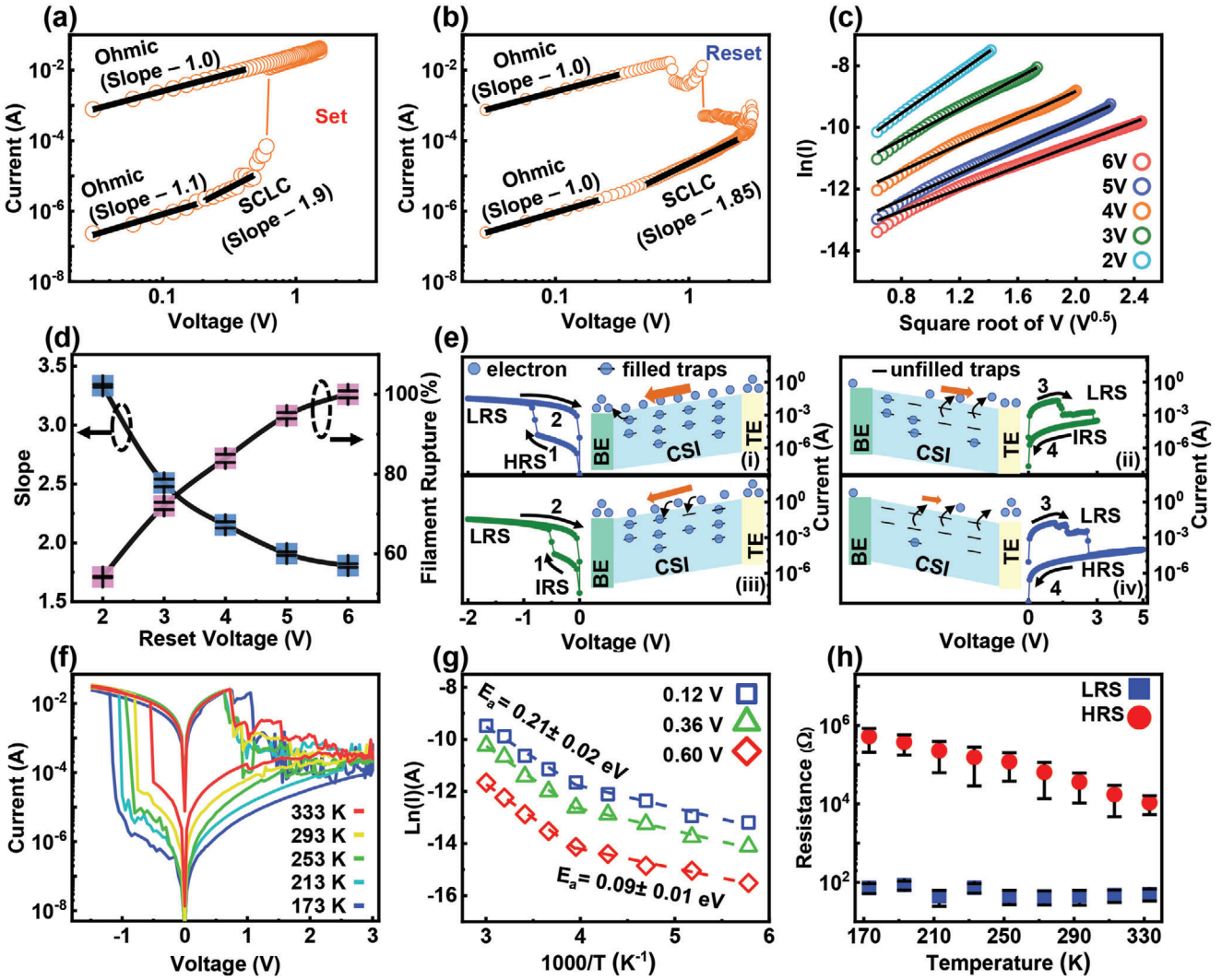


Figure 3. Conduction mechanism and thermal resilience analysis of the CSI RRAM device. Log (I)-log(V) plots for a) set and b) reset process, complemented by linear fittings, highlighting Ohmic and SCLC regimes. c) Richardson plots (ln(I) vs square root of V) analysis of the HRS states of reset voltage-based multilevel states. d) Extracted slope and intercept values from the Richardson plots. e) The reset voltage-based multilevel resistive state mechanism is elucidated with trap state dynamics. f) Thermal resilience of RS properties of CSI device in measured temperature-range 173 to 333 K. g) Arrhenius ln(I) vs. 1000/T plots at different voltages and h) temperature-dependent HRS and LRS values at $V_{read} = \pm 0.3$ V.

1 (I ∝ V), it suggests that the conduction mechanism in LRS is governed entirely by Ohmic behavior. This is attributed to the formation of conductive filaments between the top and bottom electrodes within the CSI RS layer. Conversely, the linear fitting of the log (I)-log(V) curves for the HRS states reveals two distinct slopes, signifying two separate conduction mechanisms in the voltage range from 0 to 3 V. Within a smaller voltage range (< 0.3 V), the *I–V* characteristics adhere to Ohm Law (I ∝ V). This behavior results from thermally generated free carriers surpassing the injected carriers, as detailed by.[52]

$$J = \frac{qn\mu}{d} V \quad (1)$$

where J represents the current density, q is the electronic charge, and n is the carrier density under thermal equilibrium. Furthermore, d signifies the thickness of the film, while μ represents the mobility of charge carriers. When the voltage surpasses 0.5 V, the linear fit slope for the *I–V* curve transitions to ≈1.9. This change aligns with Child's square law ($I \propto V^2$), [53] expressed a

$$J = \frac{9}{8} q\varepsilon_r\varepsilon_0\mu\frac{V^2}{d^3} \quad (2)$$

where ϵ_r represents the relative dielectric constant, and ϵ_0 is the permittivity of free space. Within this region, the shallow trap states capture injected charge carriers, which causes the free carrier concentration to be significantly lower than the trapped charge. Consequently, electron hopping via shallow traps combined with the trap-unfilled space charge limited current (SCLC) mechanism influences the leakage current. In the SCLC process, as all traps become occupied by carriers, a rapid influx of injected

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carriers moves through the conduction band, causing an abrupt surge in current. The transition from the Ohmic to the Child's square region reflects the shift from the trap-unfilled to the trap-filled phase of SCLC. The data from the HRS region suggests that the conduction mechanism primarily hinges on the SCLC transport property. To thoroughly understand the origins of the multiple HRS states in the ITO/CSI/Ag-based RRAM devices, examining the conduction mechanism of each resistance state is imperative. As such, *I–V* curves for multiple HRS states have been replotted in the high field region (V > 0.5 V) as ln(I) versus $V^{1/2}$ on a logarithmic scale (see Figure 3c). The linearly fitted trend observed in the ln(I) versus $V^{1/2}$ pots suggests that the dominant conduction mechanism is thermionic emission, as described by the subsequent equation.[31,54]

$$J = A^{**}\ T^2 exp\left[\frac{-q\left(\varphi_B - \sqrt{\frac{qV}{4\pi\varepsilon_1 d}}\right)}{\kappa_b\ T}\right] \quad (3)$$

where A^{**} is the effective Richardson constant, φ_B is the barrier height, ϵ_i is the dielectric permittivity, and d represents the rupture length of the filament. By taking the logarithm of Equation (3), the relationship between the intercept and slope can be expressed as

$$Slope \propto \sqrt{\frac{1}{d}} \quad (4)$$

Figure 3d presents the slope values derived from Equation (4) across various reset voltages. As the RESET-stop voltage increases from 2 to 6 V, the device slope descends from 3.33 ± 0.01 to 1.81 ± 0.01 (refer to Figure 3d left y-axis). A diminished slope in the ln(I) versus $V^{1/2}$ plots implies a larger d value, as per Equation (4). This can be interpreted as an extended filament rupture length, shown in Figure 3d, right y-axis. As a result, higher RESET-stop voltages produce elevated HRS states. Notably, our empirical findings regarding the filament rupture length variation with RESET-stop voltages align well with the previously reported simulation models like the Stanford/ASU Model[55] and Huang Physical Model.[56] Furthermore, the growth/rupture of the filament is explained by the movement of iodine ions (I^-), iodine vacancy generation (V_I), and their recombination.[55,56] The temperature-dependent electrical characterization was performed to evaluate/understand defect states in CSI films (Figure S16, Supporting Information). The extracted activation energy (0.30 ± 0.02 eV; more details on activation energy are provided in the Supporting Information) confirms the presence of iodine vacancies (V_I) in the CSI films. These iodine vacancies could act as trap states and possibly contribute to the multi-bit memory characteristics of the CSI RRAM devices. The proposed schematic representation of multi-bit switching is shown in Figure 3e, which shows the fillings of trap states and electron flow at various switching processes. During the set process (0 V → −2.0 V → 0 V), all traps are filled, and the device state changes from HRS to LRS, as shown in Figure 3e-i. The reset process (0 V → 3.0 V → 0 V), where the partial filament dissolution happens and the trap states become unfilled, caused the device to switch from LRS to intermediate resistance state (IRS) (Figure 3e-ii). Again, all traps are filled during the set process (0 V → −2.0 V → 0 V), and the device condition changes from IRS to LRS (Figure 3e-iii). When the applied reset voltage is from 0 V → 5.0 V → 0 V, due to the increased electric field, filament dissolution (rupture) is more significant, and more traps are available, resulting in the high HRS (Figure 3e-iv). In summary, larger filament chunks dissolve via recombination due to higher reset voltages, resulting in a larger rupture length, which provides control over multi-bit memory.[27]

The environmental and thermal stability are key aspects of figures for evaluating RRAM devices.[57] The optical images of devices and the bipolar *I–V* characteristics of CSI devices with Au electrodes stored in ambient conditions (room temperature, relative humidity ≈20–30%) over 10 days (Figure S17 and Table S5, Supporting Information). According to the literature, the stability of devices depends on electrode materials and RS layers.[9,57–59] The slight shift in set voltage and increase in HRS state current was observed over 10 days but within the durable range in the specification for an RRAM device (Figure S18,Supporting Information). The decline in the performance of devices might be due to degradation of the electrode/CSI interface, the reduction of crystallinity, and non-radiation in the devices, which increases HRS current.[9,57–59] However, with the adoption of novel encapsulation techniques, the longevity of CSI RRAM is expected to be improved.[39,60,61] The thermal stability of CSI-based RRAMs was assessed across a temperature range of 173 to 333 K, as depicted in Figure 3f. *I–V* characteristics under measured range verified the robust bipolar RS traits, even in severe cold conditions (Figure 3f; Figure S16, Supporting Information). A noticeable change in the HRS of these devices with rising temperature (highlighted in Figure 3f) aligns with findings from previous studies on metal halide perovskite-based RRAMs.[62–64] Activation energy was deduced from the Arrhenius plot (Figure 3g) to interpret the variation in HRS values. The determined E_A values stand at ≈0.21 ± 0.02 eV for the 273–333 K range and ≈0.09 ± 0.01 eV for the 173–273 K range. Interestingly, these values correspond with previously documented E_A for iodine vacancy (V_I) and Sn interstitial (Sn_i),[21,65] suggesting the presence of both V_I and Sn_i in the CSI RS layer. A graphical representation of HRS and LRS values, taken from Figure 3f at 0.3 V, is available in Figure 3h. The HRS value declined as temperatures increased, while LRS stayed consistent (Figure 3h). This drop in HRS values can likely be attributed to the thermal activation of V_I and Sn_i in the CSI at elevated temperatures. To further verify the presence of defects in the CSI RS layer, electrical conductivity (σ) measurements were undertaken between 148 and 373 K (Figure S19a, Supporting Information). An initial gentle uptick in conductivity was observed for temperatures below 273 K, but a marked escalation in σ values was recorded (Figure S19b, Supporting Information). The activation energy, derived from the linear segment of the Arrhenius plot (Figure S19b, Supporting Information), came in at (133 ± 10) meV. Given this, it is inferred that the observed activation energy sufficiently activates V_I and Sn_i defects in CSI. This suggests that the conductivity increase with temperature is likely due to the thermal stimulation of V_I and Sn_i defects in CSI under higher temperature conditions.

In the realm of perovskite-based RRAM devices, three primary switching mechanisms are recognized: electrochemical metallization (ECM), valence change memory (VCM), and an electrode-RS layer interface-type mechanism.[66] The CSI RRAM

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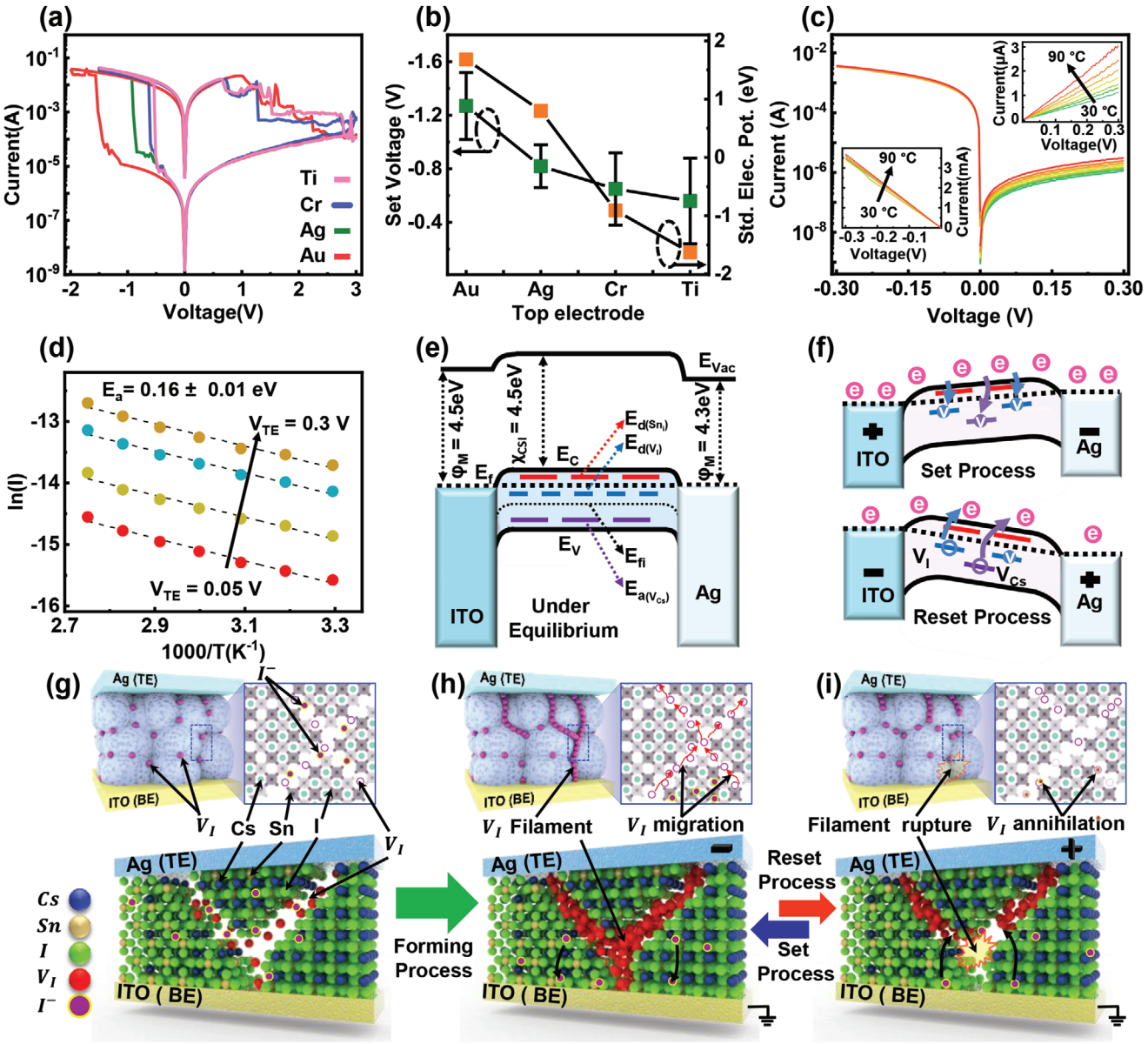


Figure 4. Resistive switching mechanism and proposed RS model in CSI RRAM and temperature-dependent HRS and LRS analysis: a) Self-compliance, bipolar I-V RS characteristic of CSI device with active and inert top electrodes. b) Variation of set voltage of CSI devices with top electrodes versus standard electrode potentials of top electrode. c) Temperature effects on HRS and LRS states of devices visualized, d) supplemented by an Arrhenius plot and activation energy derivations. The energy band diagrams of ITO/CSI/Ag devices: e) at equilibrium and f) during set/reset transitions. The schematic of the proposed switching model demonstrates the valence change memory (VCM) mechanism in the CSI RRAM devices: g) initial, h) set, and i) reset states.

devices lack Schottky diode-like rectifying *I–V* characteristics,[67] negating the electrode-RS layer interface-type switching mechanism. The influence of varying top electrode diameters (400–600 μm) on CSI RRAM devices was explored to discern the operative switching mechanism. Interestingly, this alteration in the top electrode dimension did not significantly affect the switching cycles or the On/Off ratio of the CSI devices, as illustrated in Figure S20 (Supporting Information). This indicates that interface-type switching is not prevalent in these devices. To investigate whether the switching mechanism is ECM or VCM-based, we analyzed temperature-dependent HRS and LRS measurements and inspected the effects of diverse top electrodes on RS properties. Independent of the choice of active top electrodes (i.e., Ag, Cr, or Ti) or inert ones like Au, consistent bipolar RS behavior emerged, as visualized in **Figures** 4a and S12 (Supporting Information). All these top metal electrodes exhibited a similar On/Off ratio order (≈10^3), as shown in Figures S12 and S21 (Supporting Information). Nonetheless, the Au top electrode stood out with a higher set voltage than its counterparts, a discrepancy traced back to the differing electrode reduction potentials of the metals, depicted in Figure 4b. Examining the Log(*I*)-Log (*V*) curves for LRS and HRS states at smaller voltage domains (<0.4 V) across all

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top electrode-adorned devices revealed adherence to Ohm's law (Figure S22, Supporting Information). Similarly, HRS states at elevated voltage regions aligned with the SCLC mechanism, which is speculated to be attributed to superficial trap states within the CSI RS layer (Figure S22, Supporting Information). The uniform switching attributes and consistent conduction method across devices outfitted with Cr, Ti, Au, and Ag electrodes effectively dispel the notion of an Ag filament-centric ECM mechanism. Previous studies have emphasized that a potent electric field (exceeding 10^5V cm^{-1}) is a prerequisite for forming an Ag filament during the RRAM device set process.[40,68] However, in our analysis, the electric field produced during this phase was ≈10^4 V cm^{-1} for all the top electrodes (Table S6, Supporting Information), rendering it inadequate for Ag ion migration in the thick (690 nm) CSI RS layer. Moreover, the device's negative set voltages, especially with Ag electrodes, act as deterrents, thwarting the migration of positively charged Ag cations and thus obstructing the formation of Ag conductive filaments within the devices. It is worth noting that VCM filament-driven RS has been identified in RRAMs featuring a negative set voltage and a perovskite layer thickness exceeding 200 nm, as recorded in Table S4 (Supporting Information).

To delve deeper into the CSI devices' switching mechanism, temperature-dependent HRS and LRS states were analyzed (Figure 4c). Remarkably, the resistance of the LRS remained consistent across temperatures ranging from 273 to 353 K (Figure 4c). This consistency negates the possibility of metallic filament-based RS in CSI devices. Conversely, as temperatures escalated, a discernible dip in the HRS resistance emerged, potentially arising from the thermal activation of defects in the CSI RS layer. Overall, with a robust ≈70% retention On/Off ratio, the CSI devices proved thermally reliable up to 353 K (Figure 4c). To discern the activation energy (E_A) and identify potential defects within the RS layer, an Arrhenius plot of ln(I) versus 1000/T(K) was crafted (Figure 4d). An extracted activation energy of 0.16 eV surfaced, mirroring values previously attributed to iodine vacancy (V_I) or tin interstitial (Sn_i).[65] Notably, this E_A coincides with that deduced from the device's temperature-dependent electrical conductivity (Figure S19, Supporting Information). These empirical findings underscore that the RS predominantly hinges on V_I defects in the CSI RRAM. Furthermore, we performed an XPS depth profile analysis of the CSI devices in the high-resistance state (HRS) (Figures S23 and S24, Supporting Information) and low-resistance state (LRS) (Figure S25, Supporting Information), to provide additional experimental evidence on the underlying switching mechanism. The schematic of the vertical CSI RRAM device (Ag/CSI/ITO) structure used in the XPS depth profiling analysis (Figure S23, Supporting Information). The XPS depth profiles of the CSI device in HRS (Figure S25, Supporting Information) and LRS states (Figure S25, Supporting Information) were examined to analyze the chemical bonding states of Ag (electrode) and constituent elements (Cs, Sn, I) for a deeper understanding of the resistive switching (RS) mechanisms. Ar ion etching, with a fixed Ag etching rate of ≈9 nm min^{-1}, was employed. The XPS depth profiles of Cs ($Cs3d_{5/2}$, $Cs3d_{3/2}$) and Sn ($Sn3d_{5/2}$, $Sn3d_{3/2}$) exhibited no significant transition of peak positions during Ar etching at different depths, indicating unchanged chemical states in both HRS (Figure S24b,c, Supporting Information) and LRS states (Figure S25b,c, Supporting Information). Similarly, none of the notable transition of peak positions for Ag3d ($Ag3d_{5/2}$, $Ag3d_{3/2}$) and I3d ($I3d_{5/2}$, $I3d_{3/2}$) was observed during Ar etching at different depths in the HRS state (Figure S24a,d, Supporting Information). In contrast, for the device switched to LRS, the broadening and a clear gradual shift of the peak positions ($Ag3d_{5/2}$, $Ag3d_{3/2}$, Figure S25a, Supporting Information) to lower binding energies (≈0.2 eV) was observed starting from etching time = 7 min (≈30 nm above the Ag/CSI interface), suggesting oxidization of the Ag electrode. The oxidization of Ag began to form the Ag/CSI interface and extended into the Ag electrode. The oxidized Ag signals might be attributed to the formation of AgI_x at the Ag/CSI interface, which is consistent with previously reported literature.[59,65,69] The construction of AgI_x at the interface generates more iodine vacancies (V_I) in the CSI layer. Similarly, the broadening and a clear gradual shift of the peak positions ($I3d_{5/2}$, $I3d_{3/2}$, Figure S25d, Supporting Information) to higher binding energies by ≈0.28 eV was observed starting etching from 10 to 80 min during depth profiling of I, suggesting the presence of abundant V_I in the CSI layer. The migration of the I ions occurs under electrical stress; in the meantime, V_I is eventually formed. This phenomenon indicates that the V_I CFs are attributed to switching from HRS to LRS in a CSI device. Literature suggests that these observations in the LRS state signify the formation of a V_I-based conducting filament or VCM switching mechanism in CSI devices, aligning with our experimental XPS depth profile analysis and reported findings.[65,70–72]

The physical model of conductive filaments (CFs) is elucidated in Figure 4g–i. Central to the RS mechanism is the movement of iodine ions (I^-) and the alignment of V_I.[52,73] At the outset, the CSI layer harbored a random distribution of V_I and I^- (Figure 4g). A negative potential applied to the top electrode (TE) as Ag spurs I^- ions toward the bottom electrode (BE) as ITO, generating V_I vacancies in the process. An intensifying negative potential beckons the V_I toward the top electrode, forging V_I based CFs between BE and TE (Figure 4h). This metamorphosis from the "OFF" (HRS) to "ON" (LRS) state is mirrored by a pronounced surge of current in the *I–V* curves (Figure 2a). Contrarily, a positive potential to the TE propels I^- ions to merge with the V_I vacancies, thereby intensifying the recombination. As the applied potential increases, this recombination obliterates the V_I vacancy, causing filament disintegration. This shift, embodied in Figure 4i, returns to the "OFF" (HRS) state, prompted by the dissolution of the filamentary conductive path. In essence, the CSI RS layer's RS memory dynamics in the glass/ITO/CSI/Ag memory devices are orchestrated by the creation and disintegration of V_I vacancy-based CFs

A flexible RRAM device was constructed on ITO-coated PET substrates with an ITO/CSI/Ag configuration. The schematic representation and corresponding optical image for the devices are captured in **Figure 5**a. Figures 5b and S26 (Supporting Information) show the schematic diagram of the mechanical stability testing apparatus and its optical imagery. For further visual clarity, Figure 5c presents optical images of the flexible devices at varying bending radii. To ascertain the durability and reliability of the flexible devices, over 100 direct current (DC) switching cycles were examined, as illustrated in Figure 5d. The states of LRS/HRS, extracted at a read voltage of ± 0.2 V from these cycles, are depicted in Figure 5e. Figure 5f graphically represents the statistical distribution of the extracted SET/RESET

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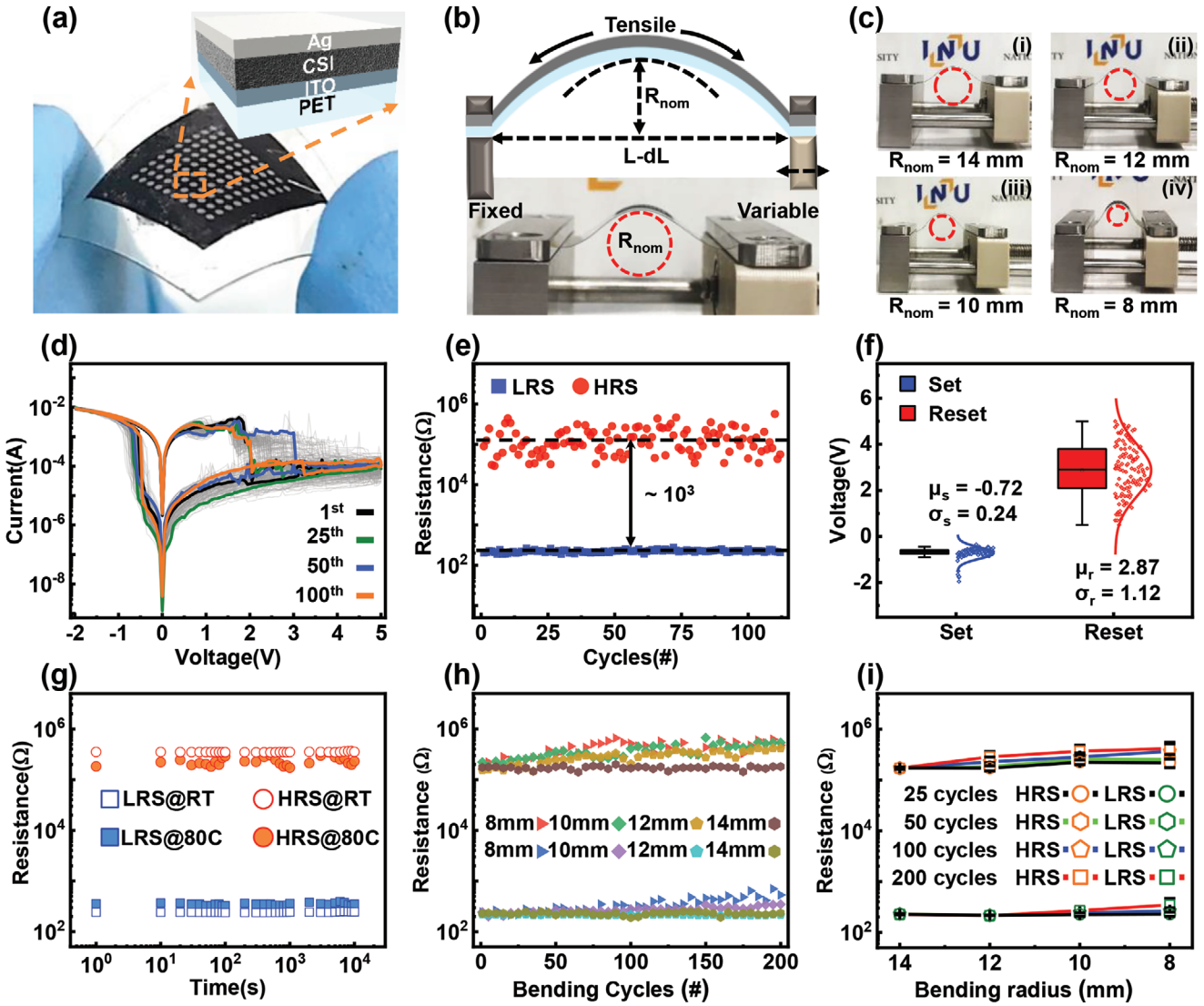


Figure 5. Mechanical flexibility of flexible PET/ITO/CSI/Ag RRAM under different bending cycles and bending radi: a) optical image of the flexible CSI RRAM device. b) schematics of mechanical bending test and an optical image of flexible RRAM. The optical images of the CSI device under different radii: c-i): 14 mm to c-iv) 8 mm. d) Endurance of the flexible device over 100 switching cycles. e) HRS and LRS value derived at $V_{read} = \pm 0.3$ V and f) statistical distribution of set/reset voltages determined from endurance analysis. g) Retention of HRS and LRS state over 10^4 s at ambient and 80 °C conditions. Analysis of h) bending resilience on HRS and LRS values over 200 bending cycles at different bending radii. i) highlighting the mean of HRS and LRS post-specific bending iterations.

voltages. Notably, the average On/Off ratio for the device stood at roughly $(6.5 \pm 4.7) \times 10^2$, with set and reset voltages amounting to $-(0.72 \pm 0.24)$V and (2.87 ± 1.12)V, respectively. The SET and RESET voltage variations were measured at 33% and 39% CVs, respectively. The experimentally observed alterations in RS characteristics between flexible and rigid devices can potentially stem from variations in surface energy, surface roughness, or the sheet resistance (150 Ω cm) of ITO-coated PET substrates.[74] Retention data for HRS and LRS (at $V_{read} = \pm 0.2$ V) spanning 10^4 s at room temperature and an elevated 80 °C are displayed in Figure 5g. Further rigorous testing gauged the stability of the devices across different bending radii (14, 12, 10, and 8 mm) over 200 bending cycles (Figure S27, Supporting Information). After every quintet of bends spanning these 200 cycles, the states of devices (i.e., HRS and LRS) were examined. The resulting resistance states for each bending radius across the cycles can be viewed in Figure 5h. Figure 5i illustrates average HRS and LRS values at each radius after specific cycle intervals (25, 50, 100, and 200). Interestingly, during the initial 50 cycles, HRS/LRS values displayed consistent stability across all radii (Figure 5h). However, post the 100-cycle mark, a noticeable degradation emerged, peaking most significantly at the 8 mm radius after 200 cycles (Figure 5h). This degradation likely originates from structural changes in the ITO or CSI films.[75–77] For a comparative perspective, Table S7 (Supporting Information) summarizes the performance metrics of various lead-free perovskite-based RRAMs. Notably, the CSI RRAMs investigated here align competitively with alternative materials and boast commendable

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mechanical resilience even after 200 bending cycles. These findings amplify the potential of RRAM devices, heralding them as promising candidates for data storage in wearable electronics.

2.3. BNN Application

Deep neural networks (DNNs) have garnered significant attention for their capability to address a myriad of real-world challenges, spanning from image recognition to natural language processing. In sectors like image recognition, natural language processing, and autonomous vehicles, DNN has established itself as a foundational technology. This surge in interest has driven numerous enterprises and academic research groups to innovate hardware accelerators explicitly tailored for neural networks. As DNN applications require exponentially growing data, a challenge arises: the communication between the main memory and processing units creates performance and energy bottlenecks. To alleviate these challenges, the binary neural network (BNN) has been introduced, utilizing solely binary precision for both weights and activation values.[78,79] Theoretically, BNN demands 32 times less memory than traditional networks with a 32-bit floating point footprint. Instead of executing 32-bit × 32-bit multiplications, BNN performs bitwise XNOR operations. With only a marginal loss in accuracy, these hardware advantages make BNNs more amenable for deployment in embedded systems like mobile applications. Emerging non-volatile memories (NVMs) like RRAM and PCRAM have been keenly explored as synaptic storage. The resistive memory crossbar array, wherein NVM devices are structured in a crossbar array structure, holds great promise as a BNN inference accelerator. This potential is realized when a weight matrix is linked to the conductance values of individual NVM devices through spike rate encoding, and an input vector corresponds to input voltages on the memory rows. Consequently, the vector-matrix multiplication result is represented by currents flowing through the column electrodes. During the inference phase of a BNN, both weights and activations employ binary precision (±1). However, during training, weights utilize full precision to ensure precise training. Thus, compared to full-precision neural networks, BNN considerably reduces the memory footprint and computational demand for inference tasks. The sign function is employed to convert both weights and activations into binary precision. From a hardware perspective, BNN operators can efficiently perform using XNOR operators.

The potential of CSI-based RRAM devices as synaptic weights in neuromorphic architectures is underscored by a comprehensive evaluation using the 32 × 32 × 3 RGB CIFAR-10 dataset. With its assortment of 50 k training and 10 k test samples, this dataset spans 10 diverse classes, capturing color images of both animals and objects, as detailed in Figure S28 (Supporting Information). The binary convolutional neural network (BCNN) at the core of this assessment integrates 6 convolutional layers and 3 fully connected (FC) layers. Impressively, it trains exclusively on binary weight values, drawing from the low and high resistance states (LRS and HRS) of the devices, and leverages a straight-through estimator for this purpose.[80] **Figure 6a** delineates the intricate arrangement of cells and input signals within the crossbar array, aligning with the input and weight representations of BCNN. Notably, a pair of cells on a shared word line (WL) symbolize binary weight values: a configuration where the upper cell reflects LRS and the lower one indicates HRS signifies a "1". Conversely, if the states are flipped, it translates to "−1". To incorporate analog-scale input signals at the hardware tier, a rate-coding approach is employed.[81]

The training efficacy of the BCNN on the CIFAR-10 test samples across epochs, contingent on weight configuration, is captured in Figure 6b. It is noteworthy that post 100 epochs, the analog weight network reaches an accuracy pinnacle of 84.55%, while its binary counterpart touches a commendable 83.73%. Such a marginal performance differential suggests that the binary weight-focused learning technique adeptly bridges the gap between binary and analog weight networks. Subsequently, the weight values harvested after the 100-epoch mark are incorporated through an off-chip learning methodology within the crossbar array.

For a granular understanding of how device state fluctuations influence overall system performance, the variance in LRS and HRS is integrated into the binary weight values. The ensuing weight distributions, corresponding to convolutional and FC layers, are illustrated in Figure 6c,d. The dual binarized states equate to a differential of ±25 mS, marking the conductance delta of the CSI RRAM device. A key observation is that LRS-induced fluctuations lead to a broadening of the weight distribution, while the HRS oscillations have a negligible footprint. This dominance of LRS fluctuations in dictating weight values stems from the fact that a single weight is represented by the conductance difference between two CSI RRAM devices. Though LRS variation does amplify the weight distribution, the pronounced difference between LRS and HRS ensures that weight deviations stay substantial, even when factoring in the measured LRS fluctuation of 11.69%. This implies that minor perturbations in the LRS of the CSI RRAM devices don't undermine the stability or robustness of the system. Conversely, given the substantial On/Off ratio ($>10^3$) of the CSI RRAM device, the HRS influence on the weight leans toward zero, leading to an almost undetectable weight drift. Hence, even with a sizable HRS variation ratio (39.71%), the weight distributions at the hardware juncture remain consistently aligned with the original configurations.

The impact of LRS and HRS fluctuations on classification accuracy is subsequently assessed, as depicted in Figure 6e,f. The variations in LRS and HRS span from 0% to 50%. Each data point on the associated color map represents the mean (μ) and standard deviation (σ) based on 30 test runs. An intriguing observation is the decline in μ, coupled with a rise in σ of accuracy as LRS variation escalates. In contrast, HRS variation exhibits a stable classification accuracy. This behavior underscores the previous point that weight binarization is significantly influenced by the differential between LRS and HRS, with LRS variations playing a more pivotal role than HRS fluctuations. Hence, for bolstering the system performance of BCNN, it is imperative to refine the LRS distribution in RRAM equipped with electrical digital switching. Remarkably, even with considerable variation ratios, only a minor dip in classification accuracy is observed due to the pronounced gap between the two states. The validated performance metrics considering the recorded LRS and HRS fluctuations reveal a μ of 83.14% (optimal at 83.73%) and a σ of 0.321% (ideally 0.00%). This underscores the resilience of BCNN to device state fluctuations when integrated with CSI devices.

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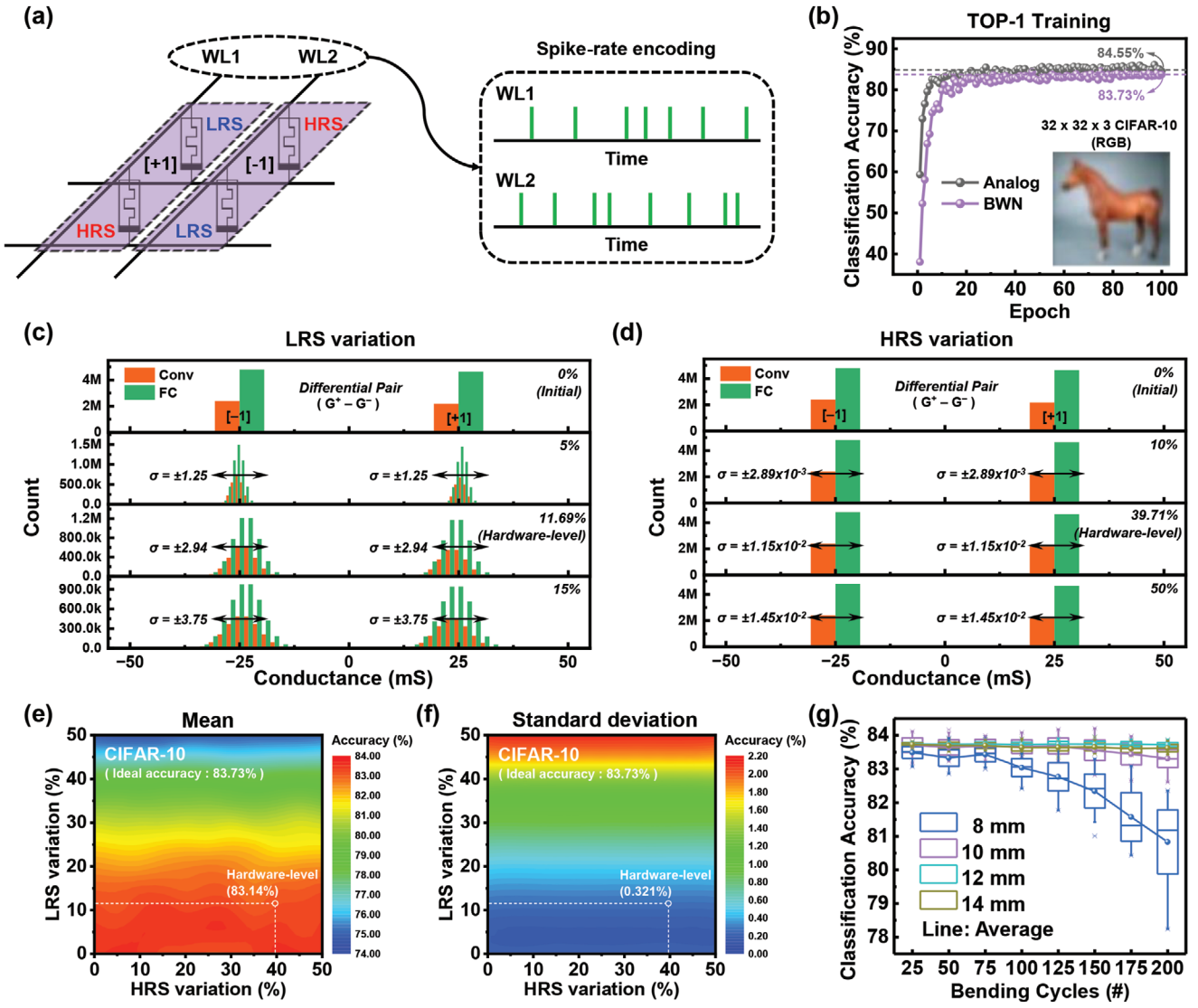


Figure 6. An illustrative introduction to the neuromorphic computing landscape via a binary convolutional neural networks (BCNNs) architecture. a) Schematic configurations representing binary weight and spike conversion encoding for analog input signals. b) Comparative training outcomes for CNNs operating with analog versus binary weight values. Binary weight distributions under the influence of c) LRS and d) HRS variations. Visual colormaps detailing e) mean accuracy and f) standard deviation metrics as functions of both LRS and HRS perturbations. g) A study of classification accuracy influenced by bending radius and the number of cycles.

Figure 6g delves into the accuracy dynamics vis-a-vis bending radius and iterative cycles. Averaging over 30 cycles, the obtained accuracies at bending radii of 8, 10, 12, and 14 mm are 83.49%, 83.71%, 83.74%, and 83.73%, respectively. Interestingly, after the initial 25 bending iterations, only minimal performance degradation is noted. However, the degradation magnitude is contingent on the bending radius as cycling progresses. At an 8 mm radius, accuracy dips to 80.83% post 200 bending cycles, whereas other radii sustain commendable performance (83.31% at 10 mm, 83.72% at 12 mm, and 83.62% at 14 mm). The pronounced degradation at 8 mm can be attributed to its substantial conductance variation, which amplifies accuracy losses as bending cycles intensify. Noteworthy is the fact that despite severe impairment in the HRS state at other bending radii, classification accuracies remain largely unperturbed. This can be ascribed to the negligible impact of HRS variations, fortified by the substantial On/Off ratio. Such findings emphasize the robustness of the binary weight network, especially in maintaining the pivotal LRS state that predominantly dictates weight configuration. Thus, the CSI RRAM device, endowed with digital electrical switching capabilities, emerges as a prime candidate for neural network implementations leveraging binary weights, in line with the proposed methodology.

3. Conclusion

The exploration into BNNs coupled with CSI RRAM devices reveals a pathway for creating compact and efficient neural

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network architectures. Their ability to replace conventional multiplication with bitwise XNOR operations stands as a testament to their optimization potential. A deeper dive into the impact of device state fluctuations on system performance outcomes further emphasizes the importance of LRS variations over HRS in dictating weight values. The overarching observation is that the ability of binary convolutional neural networks (BCNNs) to maintain a stable and robust system even in the face of potential fluctuations, especially when integrated with CSI devices, is commendable. Furthermore, this study sheds light on the flexibility and resilience of these architectures. The performance dynamics with respect to the bending radius and cycles confirm the durability of CSI RRAM devices in real-world, wearable applications. Notably, despite significant HRS variations across different bending radii, the classification accuracy rates remain largely unaffected, validating the robustness of the binary weight network. In summary, the advent of CSI RRAM devices, with their digital electrical switching capabilities, represents a pivotal shift in neuromorphic architectures. Their potential, in conjunction with BNNs, as demonstrated in this study, establishes these devices as indispensable candidates for future neuromorphic applications.

4. Experimental Section

Materials and Synthesis: Stable, environmentally friendly, lead-free CSI particles were produced using the hydrothermal method, drawing from protocols outlined in the prior work.[9] To begin, 0.3839 g of cesium acetate (2 mmol CH_3COOCs, Sigma-Aldrich, ≥99.99%) and 0.354 g of tin (IV) acetate (1 mmol $Sn(CH_3COO)_4$, Sigma-Aldrich, 99%) were introduced into 10 mL of diluted hydriodic acid (HI, Sigma-Aldrich, ≥57%). This mixture was placed in a Teflon liner and heated on a hot plate at 50 °C for 30 min, with continuous magnetic stirring. Subsequently, the Teflon liner was positioned inside an autoclave, which underwent a hydrothermal treatment at 150 °C for 2 h. Once this process was completed, the autoclave was allowed to cool naturally to ambient conditions. Dark, solid residues were isolated from the liner and underwent several washes with isopropyl alcohol (IPA) to eliminate residual by-products. The collected precipitates were then dried in a vacuum oven at 60 °C for 9 h, achieving a base pressure of ≈15 mTorr, resulting in desiccated CSI particles. These particles underwent subsequent analysis to assess their physical properties. For film fabrication, 50 mg of the aforementioned CSI particles were dispersed in 1 ml of methanol and subjected to ultrasonication for 30 min to produce a homogenized paste. This CSI paste was then applied onto rigid ITO-coated glass and flexible ITO-coated PET substrates using the drop-casting technique. To promote solvent evaporation and ensure film compaction, the coated substrates were annealed in a vacuum oven at 125 °C for 30 min, maintaining a pressure of ≈10 mTorr. These resultant films were then harnessed as RS layers to fabricate RRAM devices.

Device Fabrication: The RRAM devices were built on commercially available tin-doped indium oxide (ITO) coated glass substrates (Omniscience. Inc, 1.1 mm thick and 20 Ω sq^{-1}), which act as bottom electrodes (BE). Subsequently, the substrate was ultrasonically cleaned in acetone, isopropyl alcohol (IPA), and deionized (DI) water for 15 min and dried with N_2 gas. The cleaned substrates were further subjected to UV ozone treatment to remove impurities. Next, the 50 µL prepared CSI paste was drop-cast onto the cleaned ITO-glass substrate and annealed in a vacuum oven at 125 °C for 30 min with a pressure of ≈10 mTorr to evaporate the solvent and densify the prepared films. Finally, the top electrode (TE) of Ag (100 nm) was deposited via thermal evaporation (EB, 15NNS009), and an electrode size of 500 µm was defined using a metal shadow mask. The device fabrication process flow and the schematic diagram of the RRAM device (ITO/CSI/Ag) are shown in Figure S29 (Supporting Information) and Figure 1b, respectively. Furthermore, the RRAM devices were implemented on ITO-coated flexible PET substrate, and their electrical were examined.

Material and Device Characterizations: The crystalline phase of the synthesized CSI particles and films was characterized by X-ray diffraction (XRD, Rigaku, and Smart Lab). XRD analysis was performed in the 2-theta scan range of 20–80° using Cu K_α radiation (λ = 1.54 Å) irradiation. The synthesized particle size, shape, and chemical composition were analyzed using field emission transmission electron microscopy (FE-TEM, JEM-7210F). The surface features and thickness of prepared films were examined by field emission scanning electron microscope (FE-SEM, JEOL, JSM-7800F). The energy-dispersive X-ray spectroscopy (EDS) studied the elemental composition and mapping of the films. The chemical composition and valence states of constituent elements of prepared films were determined by X-ray photoelectron spectroscopy (XPS, PHI 5000 Versa Probe II) technique. The electrical properties were measured using a semiconductor parameter analyzer (Agilent, 4155B). Furthermore, all voltages were applied to the top electrode (Ag), and the bottom electrode (ITO) is grounded during the DC measurement. The low-temperature *I–V* characteristics were measured using a cryogenic probe station (CRX-6.5K, Lake Shore) built with a semiconductor parameter analyzer (4200-SCS, Keithley) under vacuum with a base pressure of 5 mTorr. The liquid nitrogen continuously flows from the cryostats, and a temperature controller controls the substrate temperature from 173 to 373 K.

BCNN Learning Algorithms: The off-chip learning method was employed to train binary convolutional neural networks (BCNNs) for classifying the 32 × 32 × 3 RGB CIFAR-10 datasets. During training, synapse weights underwent binarization, and post the feedforward process, the calculated loss influenced actual weight values during the backward phase, facilitating binary weight-aware learning. Each convolutional layer utilized a 3 × 3 kernel with zero-padding of size 1 and a stride of 1, ensuring the preservation of the feature map size preceding the pooling operation. The activation function employed was ReLU, with output values normalized via a log-softmax function. Training made use of the Adam optimizer-known for blending adaptive learning rate and momentum-set at a learning rate of 0.01. Cross-entropy served as the loss function. Hyperparameters featured a mini-batch size 100, batch normalization, and weight clamping to circumvent weight value divergence during training. The Adam is an optimizer that combines adaptive learning rate and momentum and can be suitable for high-dimensional image classification.[82] The initial weight distribution was set to a uniform distribution ranging from −0.3 to 0.3. The Adam optimizer, which combines adaptive learning rate and momentum, was chosen due to its suitability for high-dimensional image classification.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

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Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

binary neural network, Lead-free Cs_2SnI_6, multi-bit memory, self-compliance, thermal reliability of RRAM, at V≈−0.82 V

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